

CAR CONNECT (OOPS)

Report Generator

```
package service;

import dao.ReservationDAO;
import dao.VehicleDAO;
import entity.Reservation;
import entity.Vehicle;
import exception.DatabaseConnectionException;

import java.util.List;

public class ReportGenerator {
    private final ReservationDAO reservationDAO;
    private final VehicleDAO vehicleDAO;

    public ReportGenerator() {
        this.reservationDAO=new ReservationDAO();
        this.vehicleDAO=new VehicleDAO();
    }

    // Report for available vehicles
    public void generateAvailableVehiclesReport() {
        List<Vehicle> availableVehicles=vehicleDAO.getAvailableVehicles();
        System.out.println("\nAvailable Vehicles Report:");
        if (availableVehicles.isEmpty()) {
            System.out.println("No available vehicles.");
        } else {
            for (Vehicle vehicle:availableVehicles) {
                System.out.println(vehicle);
            }
        }
    }

    // Reservation Report
    public void generateReservationReport() {
        try {
            List<Reservation> reservations= reservationDAO.getAllReservations(); // Fetch all
            reservations
            System.out.println("\nReservation Report:");
            if (reservations.isEmpty()) {
                System.out.println("No active reservations.");
            } else {
```

```

for (Reservation reservation : reservations) {
    System.out.println(reservation);
}
}
} catch (DatabaseConnectionException e) {
    System.out.println("Error fetching reservations: " + e.getMessage());
}
}
}
}

```

Main.java

```
ReportGenerator rg=new ReportGenerator();
```

```
rg.generateAvailableVehiclesReport();
```

```
<terminated> Main [2] [Java Application] C:\Users\WELCOME\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_b4_21.0.b.v20250130-0529
Available Vehicles Report:
```

```

Vehicle ID: 1
Model: Swift
Make: Maruti
Year: 2022
Color: White
Registration Number: TN01AB1234
Availability: Available
Daily Rate: $1500.0

```

```

Vehicle ID: 2
Model: i20
Make: Hyundai
Year: 2023
Color: Blue
Registration Number: TN02CD5678
Availability: Available
Daily Rate: $1800.0

```

```

Vehicle ID: 4
Model: Thar
Make: Mahindra
Year: 2024
Color: Black
Registration Number: TN04GH2345
Availability: Available
Daily Rate: $2500.0

```

```

Vehicle ID: 5
Model: Altroz
Make: Tata
Year: 2022
Color: Silver
Registration Number: TN05IJ6789
Availability: Available
Daily Rate: $1600.0

```

```
rg.generateReservationReport();
```

<terminated> Main (2) [Java Application] C:\Users\WELCOME\p2\pool\plugins\o

☒ Database Connected Successfully!

Reservation Report:

Reservation Details:

Reservation ID: 1

Customer ID: 1

Vehicle ID: 1

Start Date: 2025-04-01 10:00:00

End Date: 2025-04-03 10:00:00

Total Cost: \$4500.0

Status: Confirmed

Reservation Details:

Reservation ID: 2

Customer ID: 2

Vehicle ID: 2

Start Date: 2025-04-05 09:00:00

End Date: 2025-04-07 09:00:00

Total Cost: \$3600.0

Status: Pending

Reservation Details:

Reservation ID: 3

Customer ID: 3

Vehicle ID: 3

Start Date: 2025-04-10 11:00:00

End Date: 2025-04-12 11:00:00

Total Cost: \$4000.0

Status: Completed

Reservation Details:

Reservation ID: 4

Customer ID: 4

Vehicle ID: 4

Start Date: 2025-04-15 08:00:00

End Date: 2025-04-17 08:00:00

Total Cost: \$5000.0

Status: Confirmed

Reservation Details:

Reservation ID: 5

Customer ID: 5

Status: Completed

Reservation Details:

Reservation ID: 4

Customer ID: 4

Vehicle ID: 4

Start Date: 2025-04-15 08:00:00

End Date: 2025-04-17 08:00:00

Total Cost: \$5000.0

Status: Confirmed

Reservation Details:

Reservation ID: 5

Customer ID: 5

Vehicle ID: 5

Start Date: 2025-04-20 12:00:00

End Date: 2025-04-22 12:00:00

Total Cost: \$3200.0

Status: Cancelled

Reservation Details:

Reservation ID: 6

Customer ID: 3

Vehicle ID: 2

Start Date: 2025-04-03 00:00:00

End Date: 2025-04-06 00:00:00

Total Cost: \$5400.0

Status: confirmed